

Wksh No. 1: Advanced data type

Example (Record):

Pseudo code	Equivalent in C language (with C++ facilities)				
Algo Example_Record Const D = 3 Type Student = Record FirstN : String LastN : String Group : Integer End Var T : Array of D Student i : Integer Begin For i ← 0 to D-1 do Read(T[i].FirstN, T[i].LastN, T[i].Group) EndFor For i ← 0 to D-1 do If T[i].Group = 6 then Write(T[i].FirstN, T[i].LastN) EndIf EndFor End	<pre>#include<iostream> using namespace std; Const int D = 3; typedef struct Student{ string FirstN; string LastN; int Group; }; main() { Student T[D]; int i; for(i=0 ; i<D ; i++) { cin >> T[i].FirstN >> T[i].LastN >> T[i].Group; } for(i=0 ; i<D ; i++) { if(T[i].Group == 6) { cout << T[i].FirstN << " " << T[i].LastN << endl; } } }</pre> For T : <table><tr><td>Ahmed Ayad 6</td><td>Halim Hassini 4</td><td>Zeinab Laib 6</td></tr></table> Screen <table><tr><td>Ahmed Ayad Zeinab Laib</td></tr></table>	Ahmed Ayad 6	Halim Hassini 4	Zeinab Laib 6	Ahmed Ayad Zeinab Laib
Ahmed Ayad 6	Halim Hassini 4	Zeinab Laib 6			
Ahmed Ayad Zeinab Laib					

Exercise 5.01:

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Algo Exo5_1
Const
    D = 3
Type
    Student = Record
        FirstName, LastName : String
        BYr, Wl, YBac : Integer
    End
Var
    T : Array of D Student
    i : Integer
Begin
    For i ← 0 to D-1 do
        Read(T[i].FirstName, T[i].LastName, T[i].BYr, T[i].Wl, T[i].YBac )
    EndFor

    For i ← 0 to D-1 do
        If T[i].Wl = 25 And T[i].BYr ≥ 2007 And T[i].YBac = 2025 then
            Write(T[i].FirstName, " ", T[i].LastName)
        EndIf
    EndFor
End

```

For T :	Ahmed	Halim	Zeinab	Screen	Halim Hassini
	Ayad	Hassini	Laib		
	2008	2006	2008		
	16	25	25		
	2025	2025	2024		

Exercise 5.03:

Algo Exo5_3

Const

D = 3

Type

Date = **Record**

Day, Year : Integer

Month : String

End

Employee = **Record**

LastName, FirstName : String

HrD : Date

Salary : Real

End

Var

T : Array of D Employee

i, A : Integer

S : Real

Begin

For i ← 0 to D-1 **do**

Read(T[i].LastName, T[i].FirstName, T[i].HrD.Day, T[i].HrD.Month, T[i].HrD.Year, T[i].Salary)

EndFor

S ← 0

For i ← 0 to D-1 **do**

If T[i].HrD.Year = A **then**

S ← S + T[i].Salary

EndIf

EndFor

Write(S)

S ← 0

For i ← 0 to D-1 **do**

If (T[i].HrD.Year = 2010 And T[i].HrD.Month = "December") Or

(T[i].HrD.Year = 2011 And (T[i].HrD.Month = "January" Or T[i].HrD.Month = "February")) **then**

S ← S + T[i].Salary

EndIf

EndFor

Write(S)

End

For T :	Ahmed	Halim	Zeinab
	Ayad	Hassini	Laib
	15	2	28
	March	January	February
	2008	2006	2008
	16	25	25
	48000	37500	50000

Screen	Zeinab Laib

Exercise 5.04 (Additional):

Algo Exo5_4

Type

Complex = **Record**
PR, PI : Real

End

Var

NC1, NC2, S, P : Complex

Begin

Read(NC1.PR, NC1.PI, NC2.PR, NC2.PI)
S.PR $\leftarrow$ NC1.PR + NC2.PR
S.PI $\leftarrow$ NC1.PI + NC2.PI
Write(S.PR, " + i× ", S.PI)

P.PR $\leftarrow$ NC1.PR* NC2.PR - NC1.PI* NC2.PI
P.PI $\leftarrow$ NC1.PR* NC2.PI + NC1.PI* NC2.PR
Write(P.PR, " + i× ", P.PI)

End

For T :	NC1:	Halim	Zeinab
	2.5	Hassini	Laib
	7	2	28
		January	February
	NC2:	2006	2008
	4	25	25
	1.25	37500	50000

Screen	6.25 + i× 8.25
	1.25 + i× 31.125

Example (enumeration):

Pseudo code	Equivalent in C language (with C++ facilities)
<u>Algo</u> Example_Record <u>Type</u> Day = (DIMANCHE, LUNDI, MARDI, MERCREDI, JEUDI, VENDREDI, SAMEDI) <u>Var</u> i : Integer <u>Begin</u> For i ← DIMANCHE to SAMEDI do Write(i) If i >= DIMANCHE and i <= JEUDI then Write("Jour de semaine") EndIf EndFor End	<pre>#include<iostream> using namespace std; typedef enum Day{Dimanche, Lundi, Mardi, Mercredi, Jeudi, Vendredi, Samedi}; main() { int i; for(i= DIMANCHE; i<SAMEDI ; i++) { cout << i << endl; if(i >= DIMANCHE and i<= JEUDI) { cout << " Jour de semaine" << endl; } } }</pre> Screen <pre>0 Jour de semaine 1 Jour de semaine 2 Jour de semaine 3 Jour de semaine 4 Jour de semaine 5 6</pre>

Exercise 5.05:

Algo Exo5_5

Const

D = 3

Type

Matiere = (ASD1, ALG1, CRI)

Var

Note : Array of D Real

Coef : Array of D Integer

i, SCoeff : Integer

Avg : Real

Begin

For i ← ASD1 **to** CRI **do**

Repeat

 Read(Note[i])

Until Note[i] ≥ 0 And Note[i] ≤ 20

Repeat

 Read(Coef[i])

Until Coef[i] > 0

EndFor

SCoeff ← 0

Avg ← 0

For i ← ASD1 **to** CRI **do**

 Avg ← Avg + Note[i] * Coef[i]

 SCoeff ← SCoeff + Coef[i]

EndFor

Avg ← Avg / SCoeff

Write("The Average is : ", Avg)

End

For Note :

11.25	12	7
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For Coef :

5	2	3
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Screen

The Average is 10.125

Exercise 5.07:

Algo Exo5_7

Const

D = 3

Type

Product = (Bougie, Disque, Pneu)

Var

Stock: Array of D Integer

Prix: Array of D Real

Achat: Array of D Integer

i : Integer

S : Real

Begin

Stock[Bougie] $\leftarrow$ 100

Stock[Pneu] $\leftarrow$ 20

Stock[Disque] $\leftarrow$ 30

Prix [Bougie] $\leftarrow$ 200.5

Prix [Pneu] $\leftarrow$ 10000

Prix [Disque] $\leftarrow$ 300.25

S $\leftarrow$ 0

For i $\leftarrow$ Bougie to Pneu **do**

Repeat

Read (Achat[i])

Until Achat[i] $\leq$ Stock[i]

Stock[i] $\leftarrow$ Stock[i] – Achat[i]

Write("The new stock of item number ", i , " is ", Stock[i])

S $\leftarrow$ S + Achat[i] * Prix[i]

EndFor

Write("The amount to pay is ", S)

End

For Achat :

4	1	0
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Screen

The new stock of item number 0 is 96
The new stock of item number 1 is 29
The new stock of item number 0 is 20
The amount to pay is 832.25

Exercise 5.08 (Additional):

Algo Exo5_8

Const

D = 5

Type

TypeV = (CAR, TRUCK, MOTORCYCLE)

Zone = (A, B, C)

Etat = (FREE, OCCUPIED)

Place = **Record**

Num : Integer

Zn : Zone

Tp : TypeV

Et : Etat

End

Var

AR: Array of D Place

i, Co, LastInd : Integer

Begin

For i $\leftarrow$ 0 to D-1 **do**

Read (AR[i].num, AR[i].Zn, AR[i].Tp, AR[i].Et)

EndFor

Co $\leftarrow$ 0

For i $\leftarrow$ 0 to D-1 **do**

If AR[i].Et = FREE **then**

Co $\leftarrow$ Co + 1

EndIf

EndFor

Write ("The number of free spaces is ", Co)

Co $\leftarrow$ 0

For i $\leftarrow$ 0 to D-1 **do**

If AR[i].Et = FREE And AR[i].Tp = CAR And AR[i].Zn = B **then**

Write(AR[i].Num)

LastInd $\leftarrow$ i

Co $\leftarrow$ Co + 1

EndIf

EndFor

If Co = 0 **then**

Write("No free spaces")

Else

AR[LastInd].Et = OCCUPIED

EndIf

End